
Exam: Simple Linear Regression*Multiple Choice (Single Select) - 4 Answer Options***1. What does simple linear regression model mean?**

- A) The relationship between three or more variables
- B) The variance within a group
- C) The relationship between two categorical variables
- D) The linear relationship between one IV and one DV

2. What does R-squared represent?

- A) The correlation coefficient
- B) The slope of the regression line
- C) The proportion of explained variance
- D) The standard error of estimate

3. Which is NOT an assumption of simple linear regression?

- A) Independence of errors
- B) Normally distributed residuals
- C) Homoscedasticity
- D) Multicollinearity

4. What does the regression coefficient b indicate?

- A) The p-value
- B) The change in Y per unit change in X
- C) The intercept
- D) The correlation between X and Y

5. What is a standardized beta coefficient?

- A) A standardized effect in SD units
- B) An unstandardized effect
- C) The correlation
- D) The coefficient of determination

6. Which plot tests homoscedasticity?

- A) Residuals-vs.-Fitted plot
- B) Boxplot
- C) Q-Q plot
- D) Histogram

7. Which is the model formula for simple linear regression?

- A) $y = b_0 + b_1x + e$
- B) $y = a + bx$
- C) $y = a + bx + cx^2$
- D) $y = b_1x_1 + b_2x_2 + e$

8. When is a coefficient statistically significant?

- A) When $p\text{-value} < \alpha$
- B) When $\beta > 0.5$
- C) When $R\text{-squared} > 0.5$
- D) When SE is large

9. What does OLS estimation minimize?

- A) The sum of residuals
- B) The coefficient b
- C) $R\text{-squared}$
- D) The sum of squared residuals

10. What does a significant F-test indicate?

- A) Residuals are normally distributed
- B) Individual coefficients are significant
- C) Heteroscedasticity is present
- D) The overall model is significant

Solutions

1: D

2: C

3: D

4: B

5: A

6: A

7: A

8: A

9: D

10: D